

Problem 1 (Review Proposition 5.2.1). Prove by induction on n that, for all integers n such that $n \geq 4$, we have the inequality

$$n^2 \leq 2^n.$$

Solution. We use induction on n .

Base case: For $n = 4$,

$$4^2 = 16 \quad \text{and} \quad 2^4 = 16,$$

so $4^2 \leq 2^4$.

Inductive step: Assume that for some integer $k \geq 4$, $k^2 \leq 2^k$.

We must show that: $(k+1)^2 \leq 2^{k+1}$.

From the inductive hypothesis, $2k^2 \leq 2^{k+1}$.

So it is enough to prove that $(k+1)^2 \leq 2k^2$. Now

$$(k+1)^2 \leq 2k^2 \iff k^2 + 2k + 1 \leq 2k^2 \iff 2k + 1 \leq k^2.$$

Since $k \geq 4$, we have

$$k^2 \geq 4k \geq 2k + 2 > 2k + 1.$$

Hence

$$(k+1)^2 \leq 2k^2 \leq 2^{k+1}.$$

This proves the inductive step.

Therefore, by induction,

$$n^2 \leq 2^n \quad \text{for all integers } n \geq 4.$$

Problem 2 (Exercise 17.6). Suppose that a_1, a_2, b_1, b_2 are positive integers such that a_1 and b_1 are coprime and a_2 and b_2 are coprime. Prove that

$$a_1 b_2 = a_2 b_1 \implies a_1 = a_2 \text{ and } b_1 = b_2.$$

Solution. Assume that

$$a_1 b_2 = a_2 b_1.$$

We will show that $a_1 = a_2$ and $b_1 = b_2$.

Since

$$a_1 b_2 = a_2 b_1,$$

it follows that a_1 divides $a_2 b_1$. Because a_1 and b_1 are coprime, Euclid's lemma implies that a_1 divides a_2 .

Similarly, from

$$a_2 b_1 = a_1 b_2,$$

we see that a_2 divides $a_1 b_2$. Because a_2 and b_2 are coprime, Euclid's lemma implies that a_2 divides a_1 .

Thus $a_1 \mid a_2$ and $a_2 \mid a_1$. Since both are positive integers, we conclude that $a_1 = a_2$.

Substituting into the original equation gives

$$a_1 b_2 = a_1 b_1.$$

Since $a_1 > 0$, we may cancel a_1 and obtain $b_2 = b_1$.

Hence

$$a_1 = a_2 \quad \text{and} \quad b_1 = b_2.$$

Problem 3 (Set theory practice). Let A, B, C be sets. Suppose

$$A \cup C \subseteq B \cup C \quad \text{and} \quad A \cap C \subseteq B \cap C.$$

Prove that $A \subseteq B$.

Solution. We must show that every element of A belongs to B .

Let $x \in A$. We consider two cases.

Case 1: $x \in C$.

Then $x \in A \cap C$. By the hypothesis

$$A \cap C \subseteq B \cap C,$$

we obtain $x \in B \cap C$, and therefore $x \in B$.

Case 2: $x \notin C$.

Since $x \in A$, certainly $x \in A \cup C$. By the hypothesis

$$A \cup C \subseteq B \cup C,$$

we obtain $x \in B \cup C$. But $x \notin C$, so it must be that $x \in B$.

In either case, $x \in B$. Therefore every element of A belongs to B , so

$$A \subseteq B.$$

Problem 4 (Surjection practice). Let Y be a set and let $f : \mathbb{Z} \rightarrow Y$ be a function satisfying the following properties:

1. f is a surjection,
2. $f(2n) = f(2n+1)$ for all $n \in \mathbb{Z}$.

Let $E \subseteq \mathbb{Z}$ be the set of even integers. Define $g : E \rightarrow Y$, where $g(k) = f(k)$ for $k \in E$. Prove that g is a surjection.

Solution. We must show that every element of Y is the image under g of some element of E .

Let $y \in Y$. Since $f : \mathbb{Z} \rightarrow Y$ is a surjection, there exists some $m \in \mathbb{Z}$ such that

$$f(m) = y.$$

Now either m is even or m is odd.

Case 1: m is even.

Then $m \in E$, and by definition of g ,

$$g(m) = f(m) = y.$$

Case 2: m is odd.

Then $m = 2n + 1$ for some $n \in \mathbb{Z}$. By property (2),

$$f(2n) = f(2n+1) = f(m) = y.$$

Since $2n$ is even, $2n \in E$, and by definition of g ,

$$g(2n) = f(2n) = y.$$

In either case, there exists $k \in E$ such that $g(k) = y$. Therefore g is a surjection.

Problem 5 (Review Quiz 2, Question 2). Let $f : X \rightarrow Y$ be an injection. Suppose that a subset B of Y does not intersect the image of f , that is, there does not exist $x \in X$ such that $f(x) \in B$. Prove that there exists an injection

$$g : X \rightarrow Y - B.$$

In particular:

1. define g ,
2. justify that g is a well-defined function,
3. show that g is an injection.

Solution. Define

$$g : X \rightarrow Y - B \quad \text{by} \quad g(x) = f(x) \quad \text{for all } x \in X.$$

We first show that g is well-defined. Since f is a function, for each $x \in X$ the value $f(x)$ is uniquely determined, so $g(x)$ is uniquely determined as well. Also, by hypothesis there does not exist $x \in X$ such that $f(x) \in B$. Hence for every $x \in X$,

$$f(x) \notin B.$$

Because $f(x) \in Y$, it follows that

$$f(x) \in Y - B.$$

Thus $g(x) = f(x) \in Y - B$ for every $x \in X$, so g is indeed a function from X to $Y - B$.

Now we show that g is injective. Suppose $x_1, x_2 \in X$ and

$$g(x_1) = g(x_2).$$

By definition of g , this means

$$f(x_1) = f(x_2).$$

Since f is injective, we conclude that

$$x_1 = x_2.$$

Therefore g is an injection.

Hence there exists an injection

$$g : X \rightarrow Y - B.$$